

FILIPPO MARCANTONI

Robotics Engineer | Computer Vision, Robot Learning & Physical AI
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SUMMARY

Robotics Engineer with an M.S. in Robotics Engineering from Worcester Polytechnic Institute and experience across computer vision, robot learning, generative AI, swarm robotics, VR teleoperation, and industrial automation. Seeking full-time robotics software, autonomy, or embodied AI roles starting July 2026 through OPT, focused on building full-stack robotic systems that integrate perception, learning, control, hardware, and software.

EDUCATION

Worcester Polytechnic Institute (WPI)

Worcester, MA, USA

M.S. Robotics Engineering | GPA 3.90

Aug 2024 – May 2026

B.S. Robotics Engineering | GPA 3.89, Summa Cum Laude

Aug 2021 – May 2025

Coursework: ML for Robotics, Computer Vision, Robot Dynamics & Control, Swarm Intelligence, HRI, Embedded Systems.

EXPERIENCE

AI & Industrial Robotics R&D Intern | B&R Industrial Automation, Austria

Jun 2025 – Aug 2025

- Built an LLM-based code-generation copilot for B&R Automation Studio using Transformers, RAG, and Azure AI retrieval to support industrial robot programming workflows.
- Designed a robotics-specific retrieval backend with domain-aware metadata and syntax-aware chunking for Structured Text and Motion ST code, improving context relevance across large automation code corpora.

Machine Vision & AI R&D Intern | Makro Labelling SRL, Italy

May 2024 – Aug 2024

- Developed C++ and Python machine-vision pipelines for bottle orientation, label alignment, and defect inspection on high-speed labeling lines, combining segmentation, feature matching, and homography estimation.
- Benchmarked classical and deep-learning models across bottle formats, lighting, and label shifts to characterize failure modes for automated quality control.

RESEARCH

STL-to-Fluoroscopy 2D–3D Registration | FuTURE Lab, WPI

Aug 2025 – Dec 2025

- Designed a self-supervised fluoroscopy-to-volume registration workflow using DiffDRR, 3D Slicer, and DICOM processing for image-guided endovascular procedures.
- Validated patient-specific pose estimation on a 3D-printed aorta phantom without ground-truth labels.

PIELo: Reactive Swarm Programming Language (MQP) | NEST Lab, WPI

Aug 2024 – May 2025

- Designed PIELo, a reactive swarm-programming language with consensus and virtual stigmergy as first-class primitives.
- Validated distributed swarm behaviors in ARGoS3 and on Khepera IV robots with Vicon tracking.

VR-Based Multimodal Teleoperation Interface (Capstone) | HiRo Lab, WPI

Jan 2025 – May 2025

- Built a Unity + ROS teleoperation system for a robotic nursing assistant with Meta Quest head-motion mirroring, real-time multi-camera streaming, Whisper voice commands, and ZED-Mini object-detection-driven camera selection.
- Integrated VR viewpoint control, speech-based camera switching, and perception-driven camera selection to improve operator situational awareness during remote manipulation.

SELECTED ROBOTICS & AI PROJECTS

Robot Learning – Reinforcement & Imitation Learning (RBE 577 – Machine Learning for Robotics, Spring 2026)

- Implemented policy iteration, REINFORCE, A2C, and A3C for LunarLander and PyBullet Kuka vision-based grasping, reaching 55% grasp success with a CNN actor-critic policy.
- Trained robomimic policies on 59 PickPlaceCan demonstrations: LSTM Behavioral Cloning raised success from 0.02 to 0.80; Diffusion Policy with low-dimensional and RGB observations further improved success to 0.90.

Classical MSCKF & Deep Visual-Inertial Odometry (RBE/CS 549 – Computer Vision, Spring 2026)

- Implemented a stereo MSCKF VIO backend with IMU propagation, camera-state cloning, stereo feature tracking, and nullspace EKF updates; evaluated on EuRoC and RPG datasets.
- Trained IMU-only, vision-only, and fused deep VIO networks on synthetic Blender trajectories using geodesic SO(3) and trajectory-composition losses.

Einstein Vision – Monocular Autonomous-Driving 3D Visualization (RBE/CS 549 – Computer Vision, Spring 2026)

- Built a Tesla-inspired monocular perception pipeline integrating Detic, YOLOv8/DETR, Mask R-CNN, ZoeDepth/MiDaS, YOLO-3D, PyMAF, and RAFT.
- Reconstructed 3D driving scenes in Blender with lanes, vehicles, pedestrians, traffic objects, orientation, motion state, and collision-risk estimation.

3D Scene Reconstruction – Classical SfM & NeRF (RBE/CS 549 – Computer Vision, Spring 2026)

- Implemented Structure-from-Motion from scratch: SIFT tracking, RANSAC, 8-point fundamental matrix, essential-matrix recovery, cheirality, PnP-RANSAC, triangulation, and bundle adjustment.
- Built a PyTorch NeRF with positional encoding, 5D MLPs, hierarchical sampling, and differentiable volume rendering.

TECHNICAL SKILLS

Programming: Python, C++, C, C#, MATLAB, Java, JavaScript, Buzz.

Robotics & Simulation: ROS/ROS2, Gazebo, MuJoCo, PyBullet, ARGoS3, robosuite, robomimic, Unity, Blender, Vicon, VR/XR, Dynamixel, ZED SDK, Automation Studio, 3D Slicer.

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, CUDA, Transformers, LLMs, RAG, Reinforcement Learning, Imitation Learning, Behavioral Cloning, Diffusion Policy.

Computer Vision & 3D Perception: OpenCV, SLAM, Visual-Inertial Odometry, MSCKF, SfM, NeRF, RANSAC, PnP, Bundle Adjustment, Homography, Object Detection, Segmentation, Monocular Depth, Optical Flow, Point Clouds.

Tools & Infrastructure: Git, Docker, Linux, Jupyter, Azure AI, AWS, Simulink, PostgreSQL, MySQL, HPC/Slurm, CI/CD.

CAD & Design: SolidWorks, Fusion 360.

Languages: Italian (native), English (fluent), Spanish (advanced).